MAT 5210 Homework 7

1. Let $L : K$ be finite Galois and let P be unramified. Show that P splits completely in L if and only if $\mathrm{Frob}_{\mathfrak{P}} = id$ for one, and hence every, $\mathfrak{P}$ above P.

2. Let $L = \mathbb{Q}(\zeta_7)$. For each nonzero congruence class of p modulo 7, determine the residue degree f and the number g of prime ideals lying above p. Hint: $f = \mathrm{ord}_7(p)$ and $fg = 6$.

3. Let $K = \mathbb{Q}(\sqrt{5})$. For an odd prime $p \neq 5$, show that

$$p \text{ splits in } K \quad \Longleftrightarrow \quad \left(\frac{5}{p}\right) = 1,$$

and interpret the two possible values of the Legendre symbol in terms of Frobenius.

4. Suppose $C_K \cong \mathbb{Z}_6$. Determine all possible degrees $[M : K]$ for $K \subseteq M \subseteq H_K$, and describe the corresponding subgroups of C_K.

5. We computed earlier that $C_{\mathbb{Q}(\sqrt{-5})} \cong \mathbb{Z}_2$. Without explicitly constructing H_K, determine $[H_K : K]$, its Galois group, and the criterion for a prime ideal P to split completely in H_K.

6. Let $L = \mathbb{Q}(i)$. For every odd positive integer n, show directly that

$$\Theta_{L/\mathbb{Q}}((n)) = \begin{cases} 1, & n \equiv 1 \pmod 4, \\ c, & n \equiv 3 \pmod 4, \end{cases}$$

where $c(i) = -i$. Explain why this shows that principal ideals need not lie in the kernel of the Artin map for a general abelian extension.

7. Prove directly that

$$C_{\mathbb{Q}}(m\infty) \cong (\mathbb{Z}/m\mathbb{Z})^*.$$

Explain why the positivity condition at the real place is needed to distinguish a from $-a$.